

Signature:

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

End-of-Module Exam: 2026 Autumn Semester

Zürcher Hochschule

für Angewandte Wissenschaften

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Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX08-WIN.XX / Data Science Introduction
Exam date / time:	20.01.2027 / 15:30 – 16:30
Duration:	60 Minutes
Total number of pages:	2
Permissible resources:	Closed book
Calculator:	no calculator
Dictionary:	Yes

General information:

- Use a black or blue pen (invisible ink is not allowed) to write your answers.
- All options of a question (i.e. A–D) can be ticked.
- ONLY tick ONE option per question in the table.
- Scoring: each correctly answered question = 1 point.
- Write your answers only on the answer sheet provided; rough work on question pages is not scored.
- Switch off your mobile phone and put it in your bag for the duration of the examination.
- If you want to write a draft, use the back of the question pages. Only the answer sheet counts.
- Please remain seated until all exams have been collected.

Marked by:	Total points:	Grade:

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Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1	100%		
		<i>(Grade on cover sheet)</i>	
Module grade (Evento)			⇐ Rounded ⇐

Marking Table

No.	Task	Points	Corrector's, initials
1	MC written Exam	30 Points	hitz
	Total	30 Points	

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Answer sheet

ONLY the answer sheet will be scored. All options of a question (i.e. A–D) can be ticked.

ONLY tick ONE option per question in this table.

Scoring: Each correctly answered question is worth **1 point**.

Example: For question 0, C should be ticked, then it would look like this.

Question	Answer	Question	Answer
1.	(A) (B) (C) (D)	16.	(A) (B) (C) (D)
2.	(A) (B) (C) (D)	17.	(A) (B) (C) (D)
3.	(A) (B) (C) (D)	18.	(A) (B) (C) (D)
4.	(A) (B) (C) (D)	19.	(A) (B) (C) (D)
5.	(A) (B) (C) (D)	20.	(A) (B) (C) (D)
6.	(A) (B) (C) (D)	21.	(A) (B) (C) (D)
7.	(A) (B) (C) (D)	22.	(A) (B) (C) (D)
8.	(A) (B) (C) (D)	23.	(A) (B) (C) (D)
9.	(A) (B) (C) (D)	24.	(A) (B) (C) (D)
10.	(A) (B) (C) (D)	25.	(A) (B) (C) (D)
11.	(A) (B) (C) (D)	26.	(A) (B) (C) (D)
12.	(A) (B) (C) (D)	27.	(A) (B) (C) (D)
13.	(A) (B) (C) (D)	28.	(A) (B) (C) (D)
14.	(A) (B) (C) (D)	29.	(A) (B) (C) (D)
15.	(A) (B) (C) (D)	30.	(A) (B) (C) (D)

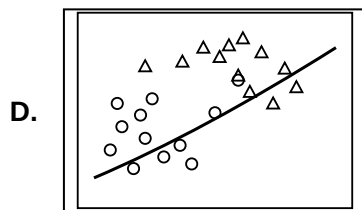
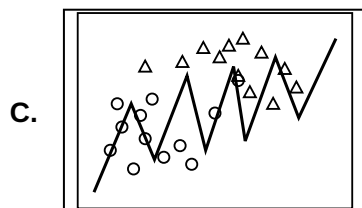
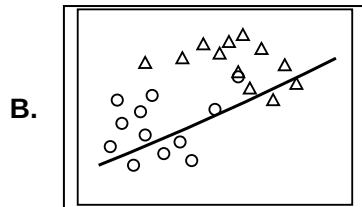
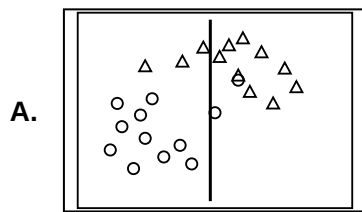
MC Data Science Introduction (30 Points)

1. Which statement about VSM and System Dynamics is NOT correct? (1)
 - A. The VSM is a recursive structure for viable systems.
 - B. System Dynamics models system behaviour over time.
 - C. Both approaches ignore feedback entirely.
 - D. They are complementary systems-thinking approaches.
2. Which statement about adaptation under Ashby's Law is NOT correct? (1)
 - A. Amplification increases the system's own variety.
 - B. If environmental variety exceeds capacity, the system automatically thrives.
 - C. Attenuation reduces or filters the environment's variety.
 - D. A system adapts by matching the environment's variety.
3. Which statement correctly distinguishes Big Data and Data Science? (1)
 - A. Big Data refers to data characteristics and infrastructure, while Data Science is a broader, transdisciplinary discipline.
 - B. Data Science is only the storage layer of Big Data.
 - C. They are synonyms.
 - D. Big Data fully contains Data Science.
4. A turbine system continuously adjusts its control parameters based on live sensor feedback. This best embodies... (1)
 - A. a one-off batch model.
 - B. unsupervised clustering only.
 - C. a closed-loop, real-time adaptive model (sense-think-act).
 - D. a static rule with no data.
5. Which single statement about overfitting versus underfitting is correct? (1)
 - A. Neither is related to the bias-variance trade-off.
 - B. Overfitting learns noise (high variance) while underfitting is too simple (high bias).
 - C. Both always reduce the training error to zero.
 - D. Underfitting reflects high variance and overfitting reflects high bias.

6. Which statement about visualization is NOT correct? (1)
- A. Dashboards help reduce data overload.
 - B. Visualizations can be biased by design choices.
 - C. A misleading axis violates objective presentation.
 - D. Visualizations are always intuitive and need no visual literacy.
7. Which statement about complexity examples is NOT correct? (1)
- A. Strong sparsity and mixed sources add to complexity.
 - B. Cause and effect are often non-obvious in complex systems.
 - C. You can always gain full overview of and control over a complex system.
 - D. Complex problems are 'messes' that require systemic thinking.
8. What does class weighting do? (1)
- A. Duplicates minority-class samples to enlarge the class.
 - B. Increases the minority class's weight in the loss function without changing the data.
 - C. Deletes majority-class examples.
 - D. Interpolates synthetic minority samples.
9. 'Actionable knowledge' in data science means... (1)
- A. raw, unprocessed data.
 - B. knowledge that is purely theoretical.
 - C. an opaque black-box output.
 - D. knowledge that is measurable and can be turned into action.
10. Which single statement about explainability is correct? (1)
- A. Explainability is irrelevant in high-stakes decisions.
 - B. Black-box opacity generally increases user trust.
 - C. Good explanations are interactive and tailored to the user's expertise.
 - D. Foundation models are inherently fully transparent.
11. Which statement about Ashby control is NOT correct? (1)
- A. The controller's variety must match the system's.
 - B. CEP and SPC are cyber-layer control functions.
 - C. Cybernetics underpins steering through feedback.
 - D. A low-variety controller reliably manages a high-variety system.
12. Which statement about the VSM is NOT correct? (1)
- A. Its subsystems monitor and regulate one another.
 - B. It is a linear plan with no recursion.
 - C. It is rooted in cybernetics.
 - D. It was used as a basis for the St. Gallen Management Model.

13. Class imbalance is dangerous mainly because... (1)
- A. it removes all features from the data.
 - B. accuracy becomes impossible to compute.
 - C. a majority-only predictor can show high accuracy while ignoring the minority class.
 - D. it makes every model too simple.
14. You have market data, but a competitor's reaction to your price cut is hard to predict. Which VUCA dimension is this? (1)
- A. Ambiguity
 - B. Uncertainty
 - C. Volatility
 - D. Complexity
15. Which single statement about unsupervised learning is correct? (1)
- A. It needs no labels, which makes the data cheaper but the results harder to evaluate.
 - B. It predicts predefined output labels.
 - C. It is the same thing as supervised learning.
 - D. It always requires a reward signal.
16. Which statement about requisite variety is NOT correct? (1)
- A. The controller must match the system's variety.
 - B. Too little variety leads to being overwhelmed by the environment.
 - C. Variety is a measure of complexity (distinguishable states).
 - D. More data automatically equals requisite variety.
17. Which statement about System Dynamics is NOT correct? (1)
- A. Causal loop and stock-flow diagrams are used.
 - B. Delays can change a system's behaviour.
 - C. It combines qualitative insight with quantitative modeling.
 - D. Reinforcing loops always stabilise systems.
18. A smart factory is best described as... (1)
- A. a CPS-based, connected production enabling real-time monitoring and control (Industry 4.0).
 - B. a manual workshop with no data.
 - C. a single spreadsheet.
 - D. an offline database of past orders.

19. Which of the following illustrations shows a classification model that suffers most from underfitting? (one answer) (1)



20. Which single statement about unsupervised algorithms is correct? (1)

- A. Fuzzy C-means forces each point into exactly one cluster.
- B. For K-means the number of clusters k is chosen in advance.
- C. K-medoids uses the mean of points as its centre.
- D. K-means requires labeled data.

21. Which statement about the target-centric approach is NOT correct? (1)

- A. It recommends starting from data and letting goals emerge later.
- B. It centres on customer value and clear priorities.
- C. It aligns activities and data with goals.
- D. It is iterative and refines insights over time.

22. Which statement about clustering and network analysis is NOT correct? (1)

- A. 'Relationships' are a building block of the 'Data DNA'.
- B. 'Network' is a recognised source of intelligence (connections, communication).
- C. Densely connected communities in a network resemble clusters.
- D. Clustering requires a fully labeled dataset to form groups.

23. Which of the following is a Big Data myth (i.e., NOT true)? (1)
- A. Veracity is an essential dimension of Big Data.
 - B. About 95% of big data is unstructured.
 - C. 'More data is always better.'
 - D. Quality and relevance matter, not just volume.
24. During EDA you find 12% missing ages and some negative prices. The best immediate action is to... (1)
- A. document and handle them (e.g., impute or clean) appropriately.
 - B. ignore it and train the model anyway.
 - C. deploy the model immediately.
 - D. delete the whole dataset.
25. Which single statement about Ashby's Law avoids the common misinterpretations? (1)
- A. More variety is always better, so just keep adding complexity.
 - B. Variety is simply the volume of data held.
 - C. The law is an exact quantitative formula, not a principle.
 - D. Variety can be balanced by amplifying one's own or attenuating the environment's variety.
26. Which statement about VUCA is NOT correct? (1)
- A. Data Science helps organisations sense, model and respond.
 - B. Uncertainty means unknown outcomes despite data.
 - C. The four dimensions all mean the same thing.
 - D. Volatility means fast, unstable change.
27. Which statement about learner types is NOT correct? (1)
- A. Lazy learners have very fast prediction time.
 - B. Lazy learners are sensitive to noise and irrelevant features.
 - C. SVMs, logistic regression and neural nets are eager learners.
 - D. Eager learners discard the training data after building a model.
28. Which of the following describes sampling bias? (1)
- A. Using a sample that is not representative of the target population.
 - B. Emphasizing data that confirms one's beliefs.
 - C. Considering only successful cases.
 - D. Over-trusting automated outputs.
29. SPC control charts that show rising process variation act as... (1)
- A. a deployment step.
 - B. an early-warning signal prompting intervention.
 - C. a clustering method.
 - D. a data swamp.

30. Which single statement about interdisciplinarity in Data Science is correct? (1)
- A. Good data science is only about algorithms.
 - B. 'Transdisciplinary' means truly integrating fields, not merely combining them.
 - C. Social science is irrelevant to data science.
 - D. Domain knowledge can be safely ignored.